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Data and text mining

Explainable multimodal machine learning model for classifying pregnancy drug safety

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Abstract

Motivation: Teratogenic drugs can cause severe fetal malformation and therefore have critical impact on the health of the fetus, yet the teratogenic risks are unknown for most approved drugs. This article proposes an explainable machine learning model for classifying pregnancy drug safety based on multimodal data and suggests an orthogonal ensemble for modeling multimodal data. To train the proposed model, we created a set of labeled drugs by processing over 100 000 textual responses collected by a large teratology information service. Structured textual information is incorporated into the model by applying clustering analysis to textual features.

Results: We report an area under the receiver operating characteristic curve (AUC) of 0.891 using cross-validation and an AUC of 0.904 for cross-expert validation. Our findings suggest the safety of two drugs during pregnancy, Varenicline and Mebeverine, and suggest that Meloxicam, an NSAID, is of higher risk; according to existing data, the safety of these three drugs during pregnancy is unknown. We also present a web-based application that enables physicians to examine a specific drug and its risk factors.

Availability and implementation: The code and data is available from https://github.com/goolig/drug_safety_pregnancy_prediction.git.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Teratogenic drugs can cause malformation of the fetus and should be avoided by pregnant women. Thalidomide, which was approved in the late 1950s as a sedative, was also used to treat nausea and vomiting during pregnancy (Dally, 1998). Thalidomide use during pregnancy caused irreversible damage to the fetus, and nearly 10 000 children were born with severe congenital malformations worldwide (Rajkumar, 2004). The Thalidomide crisis was followed by the establishment of stringent guidelines for drugs targeted at pregnant women (Sachdeva *et al.*, 2009). These guidelines enabled the relatively safe use of drugs during pregnancy, and medication use during pregnancy is common (Lupattelli *et al.*, 2014).

The widespread and growing use of drugs by pregnant women underscores the need to study the fetal risks and safety of drugs (Trønnes *et al.*, 2017), as well as the effects of drugs on both the developing fetus and the mother (Ross *et al.*, 2015). Unfortunately, the data available on pregnancy drug safety is not always consistent, and often it is difficult to correctly interpret the data that is available (Widnes and Schjott, 2008). In the absence of a central, credible and accessible knowledge bank, there is a need for advanced methods that facilitate the proper use of the data publicly available (Nörby *et al.*, 2015).

Machine learning has been successfully applied in pharmacovigilance, e.g. drug–drug interaction prediction (Shtar *et al.*, 2019), drug side effect prediction (Zhao *et al.*, 2018), and drug toxicity prediction (Vo *et al.*, 2020). Such in silico methods for pregnancy drug safety prediction have been developed (Boland *et al.*, 2017; Challa *et al.*, 2020; Marić *et al.*, 2019) and offer a promising framework for classifying the safety of drugs during pregnancy. The studies proposing those methods showed that it is possible to identify safe drugs despite the lack of clinical trials involving pregnant women, however they did not consider the relevant drug information curated prior to drug approval or integrate this information into the model. Explainability, and even causability are especially important in the field of medicine, where decision making is handled by clinicians who often feel uncomfortable with black boxes (Wang *et al.* 2020). Structured data types are preferred for when creating explainable models.

Due to the successful use of machine learning in pharmacoviligance and the ability of the multimodal data regarding drug safety to address the valued element of explainability, in this study, we aim to develop a novel explainable multimodal machine learning model for classifying pregnancy drug safety. Our proposed model has the following characteristics:

- Explainable—information that is interpretable by humans is used by the model. Molecular structure, embeddings and lower-dimensional representation are not used.
- Multimodal—information describing various aspects of drugs is combined in the model.
- Drug agnostic—the model supports the prediction of the safety of any drug, including biologics.
- Binary—the model is trained to classify drugs of 'higher risk' versus 'lower risk' for use during pregnancy.
- Premarket operation—information available before the drug is released is used by the model, thus there is no need to wait until prescription data is available before predicting pregnancy drug safety.

To tackle the task of predicting pregnancy drug safety using multimodal drug data, we suggest a number of multimodal feature extraction methods, feature selection methods and ensembles.

2 Materials and methods

Figure 1 provides an overview of the methods used end-to-end, from the data collection stage to predictions and their explanations.

2.1 Dataset

We used DrugBank 5.1.5 (Wishart *et al.*, 2018). In a multimodal dataset, samples are described from multiple perspectives; DrugBank is multimodal and textual in nature, because each feature is described using a textual name, and the features are grouped into fields, with each field covering a specific aspect of the drug. The dataset contains 10 fields, and each of the fields was used as one or more modalities in this research. This dataset contains 13 475 drugs and a total of 10 002 features, most of which are binary (Fig. 1a). Ten drug fields were used: ATC code, category, carriers, enzymes, associated condition, group, type, target, transporter, taxonomy. In addition, for each modality, we added a feature that counts the number of positive features.

Labeling drugs: Two lists of drugs labeled by domain experts were used (Fig. 1c):

- A list of drugs was created from data in the Zerifin TIS (Teratology Information Services) database. The database was used to create a list of labeled drugs. First, we gathered over 100 000 textual responses (written in Hebrew) from pharmacologists of the Shamir Medical Center to queries from doctors and members of the public regarding drug use during pregnancy and lactation. We trained a recurrent neural network to classify the

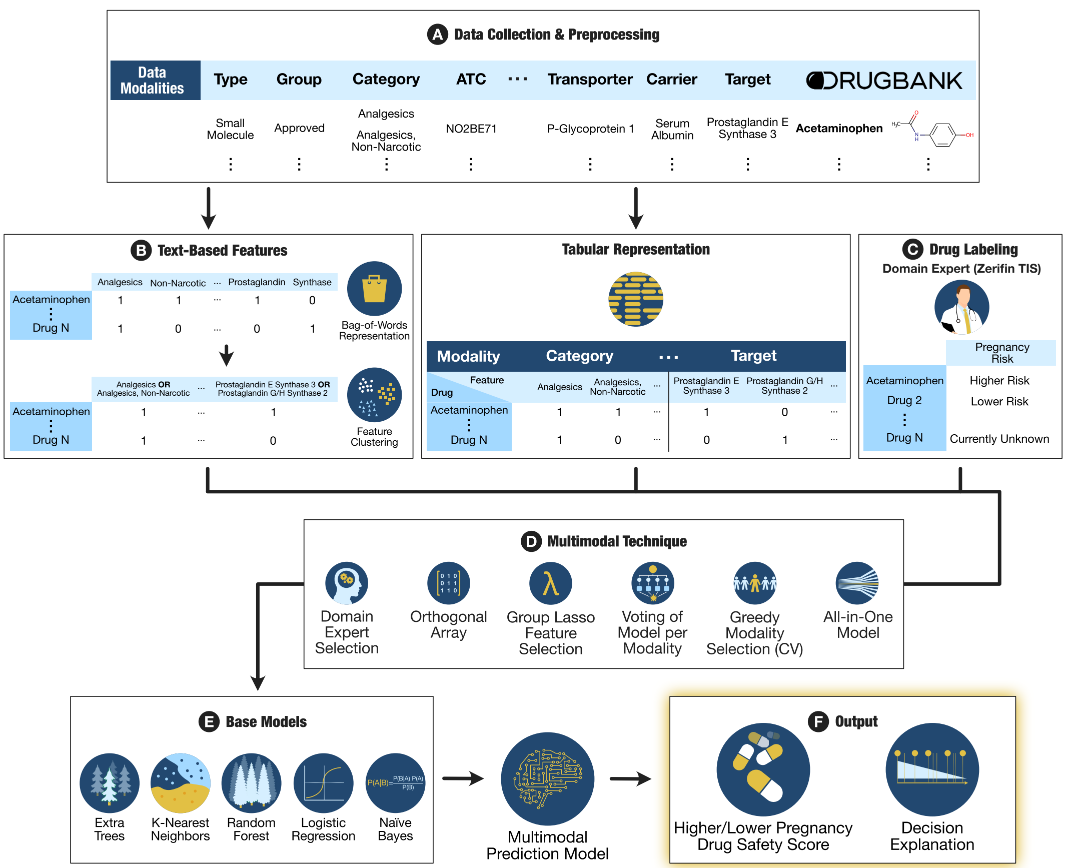

Fig. 1. Development of pregnancy drug safety prediction model. (**a**) For each drug, we collected data on 14 modalities describing the drug from DrugBank. (**b**) The textual description of each feature was used to create new text-based features: a bag-of-words (BOW) representation of the drug was created; features were clustered based on a BOW representation. (**c**) The data was labeled by domain experts. (**d**) We compared various multimodal techniques to select and combine the modalities that contribute the most to the model's accuracy. (**e**) One or more of the base models were used to create a multimodal prediction model. (**f**) The trained model was to predict the risk score. The model's decisions were explained

sentiment in each response using a manually labeled list of the 10 most queried drugs of the Shamir Medical Center: Mebendazole, Acyclovir, Omeprazole, Escitalopram, Loratadine, Nitrofurantoin, Diclofenac, Citalopram, Ibuprofen and Clotrimazole, resulting in a dataset that contains 4935 labeled samples. The score for a single drug was calculated by averaging the sentiment score assigned to each of the responses to queries regarding the drug. A pharmacologist manually validated the list and labeled each drug, assisted by the neural network; this process was repeated five times.
- The list of drugs published by Polifka *et al.* (2019). Drugs without sufficient information were not used. Drugs labeled by Poflifka *et al.* as having suggested or known risk were labeled as higher risk in the current research, and drugs labeled by Polifka *et al.* as having little to no developmental risk were labeled as lower risk here.

Sixty-two drugs were labeled by both sources. In nine cases, the label assigned by the two sources differed. Considering the information obtained by both of the sources and disregarding drugs on which they disagreed, 207 drugs in DrugBank were labeled as 'lower risk' and 126 drugs were labeled as 'higher risk'. To find each drug's associated DrugBank record, we searched for drug name in DrugBank; if it was not found there, we looked it up in Wikidata. Drugs that were not found were not used.

2.2 Processing textual features

To process the textual structured information describing the drug, we combined textual modalities selected by a domain expert. These modalities include: category, levels 2–4 of the ATC hierarchy, associated condition and taxonomy. There are a total of 6977 features. Using these features, we created two new modalities based on two types of features (Fig. 1b):

1. Bag-of-words (BOW)—each drug is described as a text document using the words describing it. The document is converted into a simplified BOW representation where the document is represented as a multi-set of its words.
2. Clustering—hierarchical clustering is used to cluster the features together based on their name, to reduce the number of features. Specifically, agglomerative clustering, which recursively merges pairs of clusters, was used. The Jaccard coefficient was used as a similarity metric, which is defined as:

$$jaccard(A,B) = \frac{A \cap B}{A \cup B}, \tag{1}$$

where A and B are features in a BOW representation. An average linkage was used as the criterion, it is defined as:

$$averageLinkage(C_1, C_2) = \frac{1}{|C_1| \cdot |C_2|} \sum\nolimits_{a \in C_1} \sum\nolimits_{b \in C_2} jaccard(a,b), \tag{2}$$

where C_1 and C_2 are clusters of features in a BOW representation. The optimal number of clusters, 3600, was found using cross-validation (CV). Finally, the new features from this process were obtained by performing a logical 'OR' operation between the features found in each cluster.

2.3 Multimodal model generation

Base model generation: To identify the best base models, we used TPOT (Olson *et al.*, 2016), a tool for automatic machine learning that uses genetic programming to identify the best base models (Fig. 1e): logistic regression, random forest (Liaw *et al.*, 2002), extra trees (Geurts *et al.*, 2006), k-nearest neighbors and naïve Bayes. The base models were used with the default parameters found in sklearn to avoid overfitting. Scaling was applied before using logistic regression, k-nearest neighbors and naive Bayes.

Multimodal technique and ensemble generation: At least one of the following multimodal techniques were used to train the multimodal models (Fig. 1d):

- All-in-one model: all of the features were used to train a single model.
- Greedy modality selection: iteratively adding the best performing modality in terms of AUC to the model and training a model using the selected modalities.
- Greedy modality elimination: iteratively removing modalities to increase the AUC of the model and training a model using the selected modalities.
- Voting of model per modality: a base model was trained for each modality separately; soft voting was performed using all of the models.
- Group lasso: group lasso (Meier *et al.*, 2008), which trains a logistic regression model and selects groups of features, was used to select modalities. The L1 regularization value, 0.01, was found using CV.
- Orthogonal array: Plackett–Burman designs (Plackett and Burman, 1946) were used to select groups of modalities. Each row in the design was used to train a base model, where each column represented a different modality. The base models were combined using soft voting. We used orthogonal arrays to reduce the number of models in the ensemble while systematically maximizing the coverage of modality combinations in the ensemble. A large version of this ensemble was also constructed, where the modalities were mixed and the design was recreated three times, each time with the three best performing models: extra trees, logistic regression and naive Bayes.
- Domain expert selection: after creating an initial model and explaining it, a pharmacologist observed the results and explanations, and selected a set of promising modalities.

These multimodal techniques can be used with any prediction model. We implemented each of the possible combinations with the base models.

Finding optimal cutting point: We found the optimal cut-off point by maximizing the F_1 score of the classifier using CV. The F_1 score measures the model accuracy by considering both the precision and recall. We report an optimal cut-off point of 32%, and the results presented throughout the article are based on this threshold value.

Decision explanation: SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017), an approach based on game theory for explaining the output of machine learning model, was used to explain the model. Specifically, we used the tree explainer, which makes it relatively fast to compute the SHAP values. To obtain stable SHAP values for the features, we repeated our experiments 30 times.

2.4 Evaluation

The main metric used in this study was the **AUC**: area under the receiver operating characteristic curve. We also report the following metrics: **AUPR**: area under the precision–recall curve, $F_1 score$, **sensitivity** and **specificity**.

Two evaluation schemes were used: **10-fold cross-validation (CV)**: data from both experts were used. **Cross-expert**: the models were trained on the drugs labeled by Polifka *et al.* and tested on the labeled set from Zerifin TIS. Each trial was repeated 30 times.

As a baseline, we implemented the model proposed by Challa *et al.* (2020) in Python. The Morgan fingerprints and molecular operating environment information are available for 609 drugs. After combining this information with the data from both experts

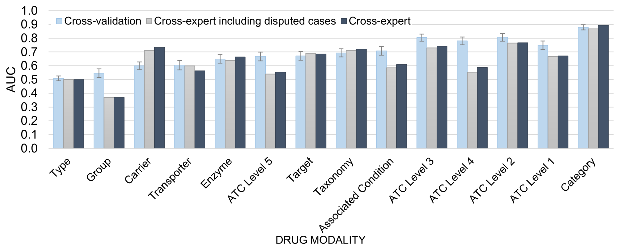

Fig. 2. Pregnancy drug safety prediction using single modalities. The performance of single modalities from DrugBank was compared. Error bars display the 95% confidence interval

we acquired during this research, 201 labeled drugs were available for CV; for training and testing in the cross-expert evaluation, respectively, 142 and 59 labeled drugs were available. Unfortunately, the teratogenic target information was not available in a structured manner, and only 64 drugs were included in the original article. Therefore, the teratogenic target information was not used.

3 Results

3.1 Pregnancy drug safety prediction with a single modality

To assess the predictive potential of each modality, we evaluated the base models with each modality, using 10-fold CV. We found that the category modality achieved the best performance, with an AUC of 0.853 over all base models; it is followed by the modalities of the level 3, 2, 4 and 1 ATC codes (Fig. 2). The best result (AUC of 0.879) was obtained by an extra trees model trained on the category modality (see Supplementary Table S1). DrugBank's category field was created by performing a manual search in PubMed; it combines the ATC codes of different levels and other manually curated drug information, and it is represented as 2773 binary variables. This initial analysis confirms that it possesses valuable information for the current task.

3.2 Multimodal pregnancy drug safety prediction

Next, we set out to determine whether additional information can improve the results. Of the 224 level 4 ATC codes found in DrugBank, 129 codes are described with a category with exactly the same name; adding these ATC codes to the categories will result in 95 new features and 129 duplicate features. Such redundancy of modalities illustrates the challenges of multimodal machine learning. To combine the different modalities in a way that exploits the unique information for maximizing accuracy, we evaluated the multimodal methods presented in Section 2.3. The best average results (an AUC of 0.889) were achieved by the large orthogonal ensemble, which was trained on all of the modalities and performs voting of a number of base models. This was followed by a large orthogonal ensemble trained on the modalities selected by the domain expert; this model, which obtained an AUC of 0.886, was followed by a voting of a logistic regression model created for each of the modalities, which obtained an AUC of 0.884 (see Table 1 and Supplementary Table S1).

To further exploit the multimodal textual nature of the dataset, we used the textual-specific methods presented in Section 2.2. Using each of these new modalities, we retrained all of the models presented earlier. We report an average AUC of 0.889 for an extra trees model trained on the clustering modality and an average AUC of 0.881 for an extra trees model trained on the BOW modality. The best results overall were achieved by a large orthogonal ensemble trained on the clustering modality added to the existing modalities. This model performs a voting of a number of the base models; in this case, an AUC of 0.891 was obtained (Table 1 and Supplementary Table S1). We found that the results of the 153 different model combinations (each model is a combination of the modalities, one or more multimodal techniques, and one or more base models) tested differed significantly from one another ($P < 1.0 \times 10^{-4}$, Friedman test). In a post hoc analysis, the best performing combination (the large orthogonal classifier trained with all of the original modalities and the clustered features) improved the results significantly ($P < 1.0 \times 10^{-4}$, paired *t*-test, Bonferroni correction) for each classifier-modality combination, except for the following: the orthogonal ensemble using all of the base features but not using the clustered features ($P > 0.05$, paired *t*-test, Bonferroni correction); only using the clustered features trained with an extra trees classifier ($P > 0.05$, paired *t*-test, Bonferroni correction); a smaller version of the orthogonal ensemble ($P > 0.05$, paired *t*-test, Bonferroni correction); and the orthogonal ensemble with the original modalities and the BOW representation ($P = 0.007$, paired *t*-test, Bonferroni correction).

3.3 Cross-expert evaluation

We repeated all of the experiments described above and used the cross-expert evaluation scheme. The experiments were performed with and without the drugs that the two sources disagreed on (those that were labeled differently by the two sources). When excluding those drugs, the best results were achieved by an extra trees model trained on the clustered features modality, which obtained an average AUC of 0.904 (see Table 1 and Supplementary Table S2—results cross-expert). The best performing single original modality was the category modality, which obtained an average AUC of 0.893 when trained on an extra trees model (Fig. 2). When including the drugs that were disagreed upon, we report the best performance by the same modality and model, which obtained an AUC of 0.888 (see Supplementary Table S3).

3.4 Comparison to models based on structural and meta-structural data

We compared the predictions made by the extra trees base model trained on the category modality to an XGBoost (Chen and Guestrin, 2016) model trained on structural and meta-structural information, as proposed by Challa *et al.* (2020). This is the only existing work that uses just information available when a drug is approved. For the 10-fold cross-validation, we report an AUC of 0.877 for the extra trees model and 0.76 for the XGBoost model proposed by Challa *et al.* (2020) ($P < 1.0 \times 10^{-4}$, paired *t*-test). For the cross-expert evaluation, we report an AUC of 0.916 for the extra trees model and 0.78 for the model proposed by Challa *et al.* (2020).

3.5 Explaining drug predictions

To pinpoint the features with a strong impact on drug safety predictions, we calculated SHAP feature importance values for all of the 2774 features used by the extra trees model trained with the category modality. We use the cross-expert evaluation scheme to interpret the model and do not include the drugs disagreed upon. We chose to provide an explanation of this model's predictions because of its simplicity and relatively high performance; it only involves raw features and an ensemble of decision trees, which are by nature easier to explain. A SHAP summary plot is presented in Figure 3. The top features for classifying a given drug as higher risk are: antineoplastic agents (ATC code L01); antineoplastic and immunomodulating agents (ATC code L); immunosuppressive agents (ATC code L04A); myelosuppressive agents; antihypertensive agents (ATC code C02); hypotensive agents; and benzene derivatives. On the other hand, the top drug features for classifying a given drug as lower risk are: anti-infective agents (ATC code S03AA); alimentary tract and metabolism (ATC code A) and anti-bacterial agents (ATC code S01AA).

Next, we set out to provide explanations for the prediction of specific drugs. First, we focus on two drugs that were classified correctly by the extra trees model trained with the category modality: Meropenem which is considered relatively safe and was assigned a risk score of 1% by the model, and Temozolomide which is considered relatively unsafe and was assigned a risk score of 99% by the

Table 1. The model's performance using cross-validation and cross-expert evaluation schemes

Modalities	Model	Cross-validation	Cross-expert
All; Clusters	Large orthogonal ensemble	**0.891 ± 0.05**	0.875 ± 0.001
All	Large orthogonal ensemble	0.889 ± 0.05	0.86 ± 0.001
All; Text	Large orthogonal ensemble	0.889 ± 0.05	0.87 ± 0.001
Clusters	Extra trees	0.889 ± 0.05	**0.904 ± 0.009**
All; Clusters	Orthogonal ensemble; Extra trees	0.888 ± 0.05	0.885 ± 0.002
All; Clusters	Voting; LogisticRegression	0.887 ± 0.06	0.866 ± 0
All; Text	Voting; LogisticRegression	0.886 ± 0.06	0.86 ± 0
Domain expert	Large orthogonal ensemble	0.886 ± 0.05	0.87 ± 0.001
All; Text	Orthogonal ensemble; Extra trees	0.885 ± 0.05	0.871 ± 0.004
All	Voting; LogisticRegression	0.884 ± 0.06	0.86 ± 0
All; Clusters	Voting; Extra trees	0.884 ± 0.05	0.823 ± 0.004
Domain expert	MultinomialNB	0.884 ± 0.05	0.826 ± 0
All; Clusters	Extra trees	0.883 ± 0.05	0.891 ± 0.01
All	Orthogonal ensemble; Extra trees	0.883 ± 0.05	0.865 ± 0.004
All; Clusters	Voting; RandomForest	0.882 ± 0.05	0.806 ± 0.006
All; Text	Voting; Extra trees	0.882 ± 0.05	0.815 ± 0.003
All; Clusters	Orthogonal ensemble; RandomForest	0.882 ± 0.05	0.859 ± 0.005
All; Clusters	Voting; MultinomialNB	0.881 ± 0.05	0.858 ± 0
All; Clusters	Orthogonal ensemble; MultinomialNB	0.881 ± 0.05	0.832 ± 0

Note: Averages and standard deviations are reported for the area under the receiver operating characteristic curve (AUC). (Bold: best score)

model. Both drugs were classified by Zerifin TIS but not by Polifka *et al.* (2019), and therefore they were not part of the training set for the model mentioned above. The SHAP decision plot explaining the decision for Meropenem is presented in Figure 4a. According to the plot, the main features contributing to the prediction are anti-bacteria agents (ATC code S01AA), lactam, beta-lactam, anti-infective agents (ATC code S03AA); amids (ATC N01BB) and antibacterials for systemic use (ATC code J01); all features have a positive value. Figure 4b presents the SHAP decision plot explaining the model's decision for Temozolomide, which is predicted to be of higher risk. The top features contributing to this prediction are antineoplastic agents (ATC code L01); antineoplastic and immunomodulating agents (ATC code L); myelosuppressive agents; immunosuppressive agents (L04A); toxic actions (a harmful effect on living organisms); and noxae; all features have a positive value.

Atorvastatin, a lipid-lowering drug, was classified as higher risk with a score of 34%; the optimized cut-off point we report is 32%, meaning that the drug was predicted to be of higher risk with very low confidence. Atorvastatin was labeled as higher risk by Zerifin TIS and as category X by the FDA. For many years, statins were considered higher risk drugs during pregnancy (Smith and Costantine, 2020) for the fear of teratogenicity, however accumulating evidence shows that it is likely that the drugs are safe for use during pregnancy (Smith and Costantine, 2020; Taguchi *et al.*, 2008; Winterfeld *et al.*, 2013). The SHAP decision plot explaining the model's prediction for Atorvastatin (see Supplementary Fig. S1a) shows that the most important features contributing to the model's prediction include toxic actions (increasing the risk); OATP1B1/SLCO1B1 inhibitors (decreasing the risk); anti-infective agents (not ATC code S03AA—increasing the risk); UGT1A3 substrates (increasing the risk) and antineoplastic agents (not ATC code L01—increasing the risk).

Next, we analyze the model's decisions for Tramadol. This drug was predicted to be of higher risk with a score of 88%, which means the drug is predicted to be relatively unsafe. However, Tramadol was labeled as lower risk by Zerifin TIS. The main features contributing to the model's prediction are NMDA receptor antagonists; cycloparaffins; opiate agonists; narcotics; opioid agonist; and opioids (ATC code N02A); all features have a positive value (see Supplementary Fig. S1b). This explanation allowed us to further investigate the incorrect prediction made by the model. To do so, we focused on the four leading features we reported for Tramadol. Four different drugs were included in each of the seven reported categories, and all of these drugs were labeled as higher risk; no drugs labeled as lower risk were included in the four categories.

We also analyzed the incorrect predictions made by the model in the cross-expert evaluation. Only one drug was defined as false negative: Silver Sulfadiazine, classified by Zerifin TIS as 'dosage dependent' (interpreted as higher risk here), was assigned by the model to have a risk score of 19.2%. This score is below the threshold of 32%, which means the drug was predicted to have a lower risk by the model. The most influential factor for this decision is the fact that the drug is an anti-bacterial agent (Supplementary Fig. S2). Thirty drugs were defined as false positive: they were classified as 'lower risk' by Zerifin TIS, however they were assigned a risk score higher than 32% by the model (predicted to be of higher risk). Six of these 30 drugs were classified by Polifka *et al.* (2019) as 'insufficient Information', a classification for drugs with no clear consensus. Twelve of the 30 false-positive drugs were assigned a risk score lower than 50% and can be considered borderline cases. We manually analyzed the remaining 12 false-positive drugs to better understand the model's behavior; the results of our analysis are as follows: (i) Clobetasol Propionate, Antipyrine and Fluorometholone: these drugs are mainly used for topical applications; (ii) Aripiprazole, Perphenazine, Colchicine, Quetiapine, Infliximab, Adalimumab and Tramadol: although these drugs are considered lower risk by the domain expert, the use of these medications warrants a discussion with the patient regarding the underlying condition and the administration of specific instructions approaching delivery or during a newborn's first months of life; (iii) Eletriptan: the safety of Eletriptan depends on the frequency of its use during pregnancy; and (iv) Diltiazem: this drug was classified by the model to be of higher risk, mainly due to its being an antihypertensive agent.

3.6 Classifying drugs of unknown safety

The web-based version of our models (Supplementary Fig. S3) includes the predictions and experts' labels for every drug in DrugBank (13 475 drugs as of version 5.1.5). Two versions of the models are included: the model used in our cross-expert evaluation which was trained on the data from Polifka *et al.* (2019), and a model which was trained on labeled data from Polifka *et al.* (2019) and Zerifin TIS (all data version). The website also includes, when available, the FDA's five letter risk categories obtained from Wikipedia and SafeFetus.com, and the Australian drug safety categorization obtained from Wikipedia. The SHAP decision and force plots of each drug are also presented on the website.

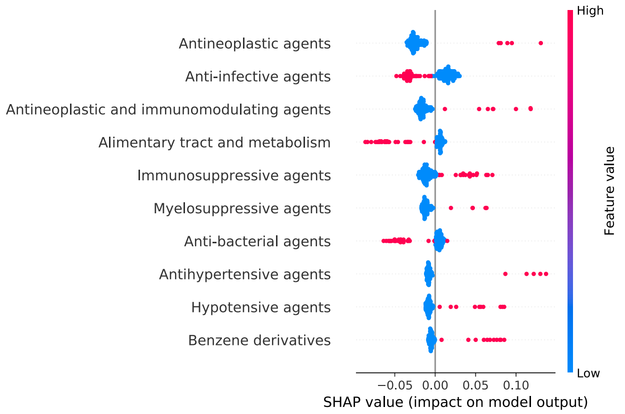

Fig. 3. Contributing features for increasing or decreasing the pregnancy drug risk score. A SHAP summary plot is used to analyze the most influential features for a model's decision. The model analyzed is an extra trees model trained with over 2700 binary features (drug categories) which were manually curated. The features are ordered in descending order based on the magnitude (importance) of the SHAP value. Each point represents an instance from the test set. The SHAP values show the distribution of the impact that each feature has on the model's output (risk score); instances with a higher SHAP value indicate a higher risk predicted by the model

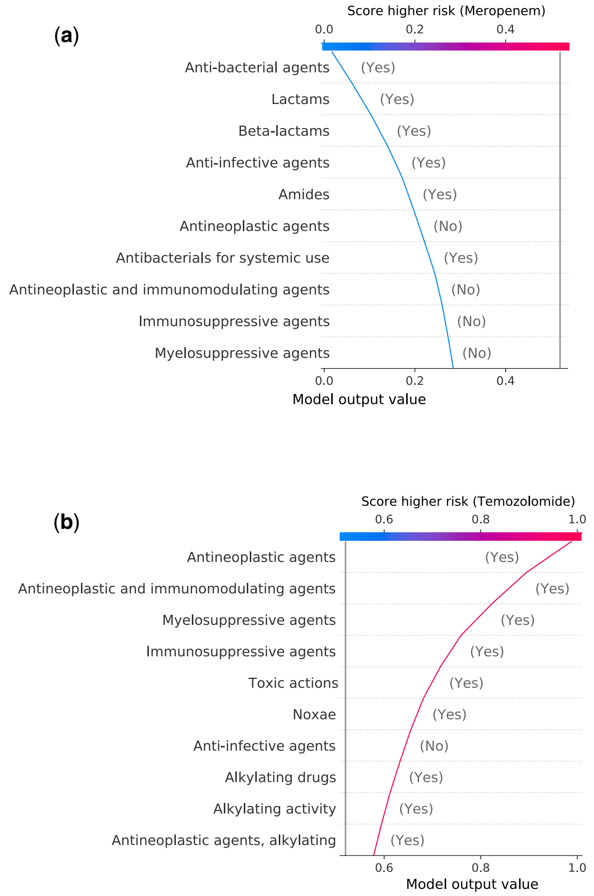

Fig. 4. Explaining the prediction of correctly classified drugs. We used SHAP decision plots to explain the predictions of the following drugs: (**a**) Meropenem; (**b**) Temozolmide. The plots show the features affecting the decision (descending order); features appearing higher in the plot indicate greater impact on the decision. Each feature can increase the risk (decision moves right) or decrease the risk (decision moves left); sharper angles indicate greater effect on the model's output (risk score). The model's base value is indicated by a gray vertical line

We used our web-based application (all data model) to analyze the risk score assigned to drugs of unknown safety. There are a total of 381 drugs that were classified as category C by at least one health organization and were not classified by the experts *(*Zerifin TIS or Polifka *et al.* (2019)). Of these, we focused specifically on drugs that were classified as category C in all of the health organization data available to us, based on the assumption that they are of greater interest to pregnant women. A domain expert from Zerifin TIS manually examined the list of 25 drugs and identified three drugs of interest: Tolbutamide, Glipizide and Meloxicam. Tolbutamide and Glipizide are drugs used to treat non-insulin-dependent diabetes mellitus (NIDDM) from the sulfonylurea class. Our model predicted Tolbutamide and Glipizide to be safe and assigned them the risk scores of 13.4% and 14%, respectively. The features contributing most to the predictions for the two drugs are: sulfones (increasing the risk); antineoplastic agents (not ATC code L01—decreasing the risk); and alimentary tract and metabolism (ATC code A—increasing the risk). Another drug identified by the experts is Meloxicam, a non-steroidal anti-inflammatory drug (NSAID) that was predicted to be unsafe according to our website's multimodal model, with a risk score of 97.5%; the three top features contributing to themodel's prediction are: cyclooxygenase inhibitors, anti-inflammatory and antirheumatic products (ATC code M01) and musculoskeletal system agents (ATC code M), all of which increase the risk. No evidence is available regarding the safety of Meloxicam during pregnancy on humans or animals, however recently, the FDA recommended avoiding NSAIDs after week 20 (https://www.fda.gov/safety/medical-product-safety-information/nonsteroidal-anti-inflammatory-drugs-nsaids-drug-safety-communication-avoid-use-nsaids-pregnancy-20). Our results support the growing concern regarding NSAID use during pregnancy.

As part of this analysis, we also focused on drugs classified by Polifka *et al.* (2019) as having 'insufficient information', which were not classified by Zerifin TIS experts and were assigned very high- or low-risk scores by the model. There are 108 such drugs; the two drugs with the highest-risk score calculated by our model are Anastrozole and Podophyllin. Anastrozole, a drug used to treat breast cancer, was predicted to be of higher risk with a risk score of 85.6%. The main contributing features are antineoplastic agents (ATC code L01), and antineoplastic and immunomodulating agents (ATC code L). Podophyllin, a drug generally administered topically to treat genital warts, was classified as higher risk with a risk score of 69.9%. The main contributing features are antineoplastic agents (ATC code L01) and noxae. Anastrozole and Podophyllin were classified as category X by the FDA, and animal reproduction studies have shown an adverse effect on the fetus, however, this information was not available to our model. Focusing on drugs which were predicted to be of lower risk, our model calculated a risk score of 0% for Darbepoetin alfa and Amodiaquine. Amodiaquine, a medication used to treat malaria, was classified as category C by the FDA, which means that animal reproduction studies have shown an adverse effect, but there are no adequate studies in humans. Darbepoetin alfa was classified as 'Compatible—Maternal Benefit ≫ Embryo–Fetal Risk' by Shanks (2015), which means that the drug might have potential fetal harm, but the benefits of using the drug are greater than the risk.

We also manually validated several drugs classified by Zerifin TIS experts as having 'insufficient information', which were not classified by Polifka *et al.* (2019) Varenicline is a smoking cessation drug used for short periods of time; it is not uncommon for women to become pregnant while using it. Varenicline was predicted to be safe according to our website's multimodal model, with a risk score of 2%; these results are in line with the new recommendation by Zerifin TIS (the recommendation has changed over time) which states that the drug is not recommended for use during pregnancy, however, in cases in which a woman becomes pregnant while using it, Varenicline is not likely to increase the risk above the general population's risk for major fetal malformation, as demonstrated in recent studies (Richardson *et al.*, 2017; Tran *et al.*, 2020). The abovementioned explanation regarding Varenicline also applies to Mebeverine, that was also classified to be safe according to our website's multimodal model, with a risk score of 2%. Mebeverine was reportedly not teratogenic in rats and rabbits, but no detailed studies are available (EMC, 2020).

Finally, we report high confidence predictions for approved drugs made by the model trained on the labeled data from both experts which were not classified by the experts: Cefmenoxime, a third-generation cephalosporin antibiotic, is predicted to be safe, with a risk score of 0% and was not classified by Shanks (2015) or any health organization agency.

Model uncertainty estimation: To examine the uncertainty of the model's prediction, we consider the variability of the predictions made by the model. We use bootstrapping to construct empirical confidence intervals. We train an ensemble of 100 models and measure the variability in each approved drug's prediction. We report an average prediction of 0.17 and an average standard deviation of 0.025 (see Supplementary Figs S4 and S5). We analyze the 10 drugs with the highest variability (Supplementary Fig. S6). Six of the drugs are beta blockers (Carteolol, Betaxolol, Metipranolol, Oxprenolol, Nadolol and Acebutolol); this is a result of interclass variability, as some of the beta blockers are safe, and some are not. Our analysis of Enoxaparin reveals that even the experts do not agree on the safety of this drug. The model learns from Heparin, a similar drug, but it is the only similar drug to learn from; according to the experts, this means that the model does not have enough similar instances to learn from, and therefore there is high variance in the prediction of Enoxaparin. Sulfacetamide is a topical drug; this drug is similar to some non-topical higher-risk drugs. According to the experts, considering the current area of interest, systemic exposure is more critical during pregnancy, and therefore fewer topical drugs were classified. This leads to fewer topical drugs in the training set, which might lead to higher prediction variability for these types of drugs.

4 Discussion

In this study, we developed explainable machine learning models for classifying pregnancy drug safety based on multimodal data. Special attention was given to integrating the relevant drug information curated prior to drug approval. Specifically, we showed that clustering structured textual drug data and giving special care to the multimodal nature of drug data can improve classification performance. A website with our multimodal models was created.

Teratology information services (TISs) are an alternative source of pregnancy drug safety information. The Zerifin TIS at the Clinical Pharmacology and Toxicology Unit at Shamir Medical Center is a telephone information service that provides information and guidance to pregnant or lactating women and healthcare professionals. The center is a member of the European Network of Teratology Information Services (ENTIS, 2020), an organization of counseling services that provide information on the safety and risks of exposure to medications and other agents during pregnancy and breastfeeding. The information collected at Zerfin TIS over the past two decades and the drugs labeled by Polifka *et al.* (2019) were used to train the machine learning models presented here.

The development of a machine learning model for drug classification begins by defining the point in the development process where predictions are required. In general, earlier in the drug lifecycle only unstructured data, such as the drug structure are available, while later in the life-cycle more structured data, such as the ATC code is available (Shtar, 2021). Besides affecting the accuracy, the data available has major implications on model explainability, because as humans, we interpret structured data better than unstructured data. Pregnancy drug safety categorization is needed mainly after drug approval. Another advantage of using structured data is the fact that it can be used to classify both small molecules and biologics. In contrast, combining the raw structure of different drug types into a single feature space is not a trivial process.

The explanation plots provided with the models and each drug prediction allow the end user to understand the decision made by the model. Some of the insights presented in the plots are well documented. Antineoplastic (Wiebe and Sipila, 1994) and immunosuppressive (Desai *et al.*, 2017) agents are associated with higher-risk drugs. Alimentary tract and metabolism agents (ATC A), anti-infective agents and anti-bacterial agents are associated with lower-risk drugs (Lacroix *et al.*, 2009). Myelosuppressive agents are known to be of higher risk during pregnancy (Delage *et al.*, 1996). Our model's explanations can aid in understanding the decisions made by the model, however, the explanations often fail to explain the underlying cause for the level of risk, which can usually be attributed to a specific toxic mechanism.

According to the results, the new modality created by clustering textual features contains critical information for the problem. This new modality combines various existing modalities and reduces the dimensionality of the data; despite this, it is still explainable. Unlike raw molecular structure and lower-dimension representations, such as PCA and embeddings, all of our models and features are explainable, however some of the features are a little difficult to interpret, such as the logical 'OR' operation between the clustered features. This illustrates the trade-off between model explainability and accuracy. Larger models, such as the large orthogonal arrays, perform better but are technically more difficult to analyze compared with decision trees or ensembles based on decision trees, which can be explained by just following the tree structure.

Our findings suggest the safety of two drugs for treating non-insulin-dependent diabetes mellitus, Tolbutamide and Glipizide and predict Meloxicam, an NSAID, to be of higher risk for use during pregnancy. These results are insufficient to advise prescribing Tolbutamide and Glipizide to pregnant women, however, about 50% of the pregnancies in the developed world are unplanned (Hohmann-Marriott, 2018), in an unplanned pregnancy a woman may not be aware that she is pregnant and may therefore inadvertently take a medication with unknown safety. Both drugs are members of the sulfonylurea class as is the drug Glibenclamide which is considered safe. Following our model's predictions for Tolbutamide and Glipizide, we can reassure a pregnant woman who has taken these drugs that the drugs are likely to be safe according to our model. In addition, our model suggests that Atorvastatin is of medium level of risk, in line with the findings of recently studies (Smith and Costantine, 2020). Finally, we classified Cefmenoxime to be of lower risk for use during pregnancy. To the best of our knowledge, this drug was not classified previously.

The proposed models and website are not meant to replace consultation with a health professional but rather to support the decisions made by patients and assist doctors in providing the correct information to patients. For example, a higher-risk drug does not mean that the drug should be avoided at all costs but rather that a decision regarding its use should be made carefully by a woman with the support of a doctor.

Our proposed models are limited to structured drug information to enable clear explanations for the predictions. Therefore, future work is needed to explore the benefits of using unstructured information, and specifically using information in the literature. In addition, to improve the prediction accuracy, our models are based on data collected before drug approval and during clinical trials, however, this means that the models cannot be used to classify drugs in the early stages of development. The higher number of false-positive cases compared with false-negative cases results from the lower threshold of 32%, which was determined by choosing a threshold that maximizes the mean of the precision and recall. This lower threshold is in alignment with a conservative and patient-safety-oriented approach. Furthermore, our proposed models are simple and binary, however, as our error analysis demonstrates, future research is needed to develop models that consider additional aspects of drug use during pregnancy, such as the administration route, dosage, pregnancy week, the risk of specific outcomes (types of fetal anomalies, etc.), indication and follow-up recommendations. These additional dimensions will improve data quality, which will require the labeling of data by domain experts, a complex task due to disagreements among the experts.

In summary, our results suggest that the safety classification of drugs used during pregnancy could benefit from combining different structured drug modalities, and specifically that textual-based features, such as the features we used in our models, are of greater value. From a translational perspective, the methods developed in this study could help health professionals identify drugs that are safe or unsafe for use during pregnancy by further analyzing our model's predictions. In addition, our insights about multimodal learning, and specifically textual structured data, can be used to tackle other machine learning problems. Some of our models are available online on our website, along with a newly published independent safety categorization of 124 drugs based on the information collected by the Zerifin TIS center. This list is also in the public domain.

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